RECONSTRUCTING TUMOR EVOLUTION USING TEMPORAL RANKINGS AND DIRECTED ACYCLIC GRAPHS TRAJECTORIES

EXAM: COMPUTATIONAL GENOMICS

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01 RESEARCH QUESTION

Question

How do tumors acquire genetic alterations across pre-defined evolutionary stages, and can we reconstruct their recurrent progression pathways through network analysis?

Main objectives

1. **Identification of key alterations & evolutionary trajectories:**
   To detect most critical genes in order to stratify patients and uncover recurrent pathways of tumor progression.
2. **Mapping mutational dependencies:**
   To reconstruct the chronological hierarchy of genetic events, revealing which early gene alterations acts as evolutionary prerequisites for later stages of cancer development.

02 DATASET AND STUDY CONTEXT

Tick Tack vs. Our Contribution

The Baseline:
Established a hierarchical Bayesian model to reconstruct the temporal order of CNA across the genome

maps genomic alterations to an abstract timeline, defining *when* they occur

Our work:
Transitioned from studying parallel timelines to building a dynamic network

uncovered recurrent mutational pathways by tracking *how* specific gene modules transition over time

Dataset overview

The dataset provides clinical-genomic data, capturing specific gene segments in different statuses across a cohort of cancer patients

- In total there are:
  - patients (samples): 1002
  - unique tumor types: 30
  - unique genes analyzed: 597
  - total segments recorded: 89696

- Key Structural Variables:
  - **sample_id**: Unique patient identifier;
  - **ttype**: Tumor type classification;
  - **gene**: The targeted gene;
  - **clock_rank**: The chronological stage (pseudo-time) assigned to that specific alteration;
  - **mutation_status**: Categorical descriptor of the segment’s status (WT, M & CI_M, CNA_driver);
  - **is_WGD**: Boolean flag indicating if the patient's genome underwent Whole-Genome Doubling.

03

COMPUTATIONAL PIPELINE & RESULTS

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A. Data extractor and scoring algorithm

Goal: To filter raw categorical mutation data and translate it into a quantitative severity score, effectively prioritizing the most heavily altered genes at each evolutionary stage

1. **Biological filtering**: Removal of all Wild-Type (WT) segments in patients exhibiting Whole-Genome Doubling (WGD);
2. **Clock filtering**: Removal of un-timed segments;
3. **Weighting function**: Application of a custom algorithm assigning a specific severity weight to each gene segment based on its mutational and genomic context;
4. **Rank-specific aggregation**: Aggregation of segment scores per gene to extract the Top 20 Genes (the most structurally impacted events) for each patient at each specific clock rank.

Weighting function specifications

Gene Scoring Logic: The score of a single segment is the product of four specific features. If a gene has multiple segments, their individual scores are summed.

1. **Mutation Type**: An exponential scale prioritizing critical structural damages:
    a. Copy number alterations (CNA_driver): 5.0;
    b. Point mutations (CI_M & M): 2.0;
    c. Wild-type (WT)=1.0;

2. **Driver Status (b_pcawg)**: Genes officially recognized as drivers by the PCAWG catalog:
    a. Cataloged genes: 2.0
    b. Non-cataloged genes: 1.0.

Weighting function specifications - Pt.2

3. **Multiplicity Weight**: Evaluates the allele dosage effect based directly on the estimated number of mutated copies.
    a. Standard karyotypes (2:0, 2:1, 2:2): dynamic multiplier calculated as 1.0 + (multiplicity × 0.2);
    b. WT and CNAs:1.0.

4. **WGD Factor (wgd_factor)**: Adjusts for global genome duplication.
    a. WGD tumors: 0.8;
    b. Non-WGD tumors:1.

Weighting function specifications - Pt.3

1. Weighting function in practice

- **Example A**: Gene with a Wild-Type status (1.0), not in PCAWG (1.0), with a balanced karyotype (1.0), in a patient with WGD (0.8).
  - Score: $1.0 \times 1.0 \times 1.0 \times 0.8 = 0.8$
- **Example B**: Gene with a CNA driver status (4.0), recognized in PCAWG (2.0), exhibiting Loss of Heterozygosity (1.5), in a non-WGD patient (1.0).
  - Score: $4.0 \times 2.0 \times 1.5 \times 1.0 = 12.0$

2. Cohort-Level Summation (Gene Weight)

These patient-level scores are then summed across the entire cohort to calculate the **global *gene_weight*** for that specific *clock_rank*. This total sum dictates which genes enter the "Top 20" ranking for the network analysis.

B. Clustering: for finding subgroups of genes

Goal: For finding subgroups of genes we need to find subgroups of patients

- Use Feature-Weighted Jaccard (based on one many genes two patients share)

$$J(U, V; \mathbf{w}) = \frac{\sum_{i \in U \cap V} w_i}{\sum_{i \in U \cup V} w_i}$$

  - Where w are the weights already described
- For grouping we used Hierarchical clustering combined with silhouette
  - Hierarchical clustering groups patients step by step based on how similar they are.
  - Silhouette scores check how well each patient fits within its group.

Clustering: for finding subgroups of genes- pt.2

Transitioning to the Gene Level: After identifying the patient clusters, we needed to shift our focus back to the genetic data.

1. Patient-to-gene substitution:
   - We replaced the clustered patients with their respective altered genes;
   - As a result, we obtained distinct **subgroups of genes** for each evolutionary stage (*clock_rank*);
2. Connecting the ranks:
   - We linked the ranks in strict chronological order, generating a Directed Acyclic Graph (DAG) that represents the timeline of the evolutionary stages

C. Connecting the subgroups: PMI

Goal: To establish and weight the actual evolutionary trajectories between gene subgroups across the ranks.

The PMI Metric:

- We applied Pointwise Mutual Information (PMI) to measure the statistical dependence between two subgroups:

$$PMI\,(A,B) = log_2\left(\frac{P\,(A)P\,(B)}{P\,(A,B)}\right)$$

- *P(A,B)* represents the observed probability of a patient transitioning from subgroup A to subgroup B.

Connecting the subgroups: PMI- pt.2

- **Interpretation & Edge Creation**:
  - PMI $>$ 0: The transition occurs more frequently than expected by chance (positive correlation)

    ↓

    An edge is drawn, representing a valid evolutionary path.

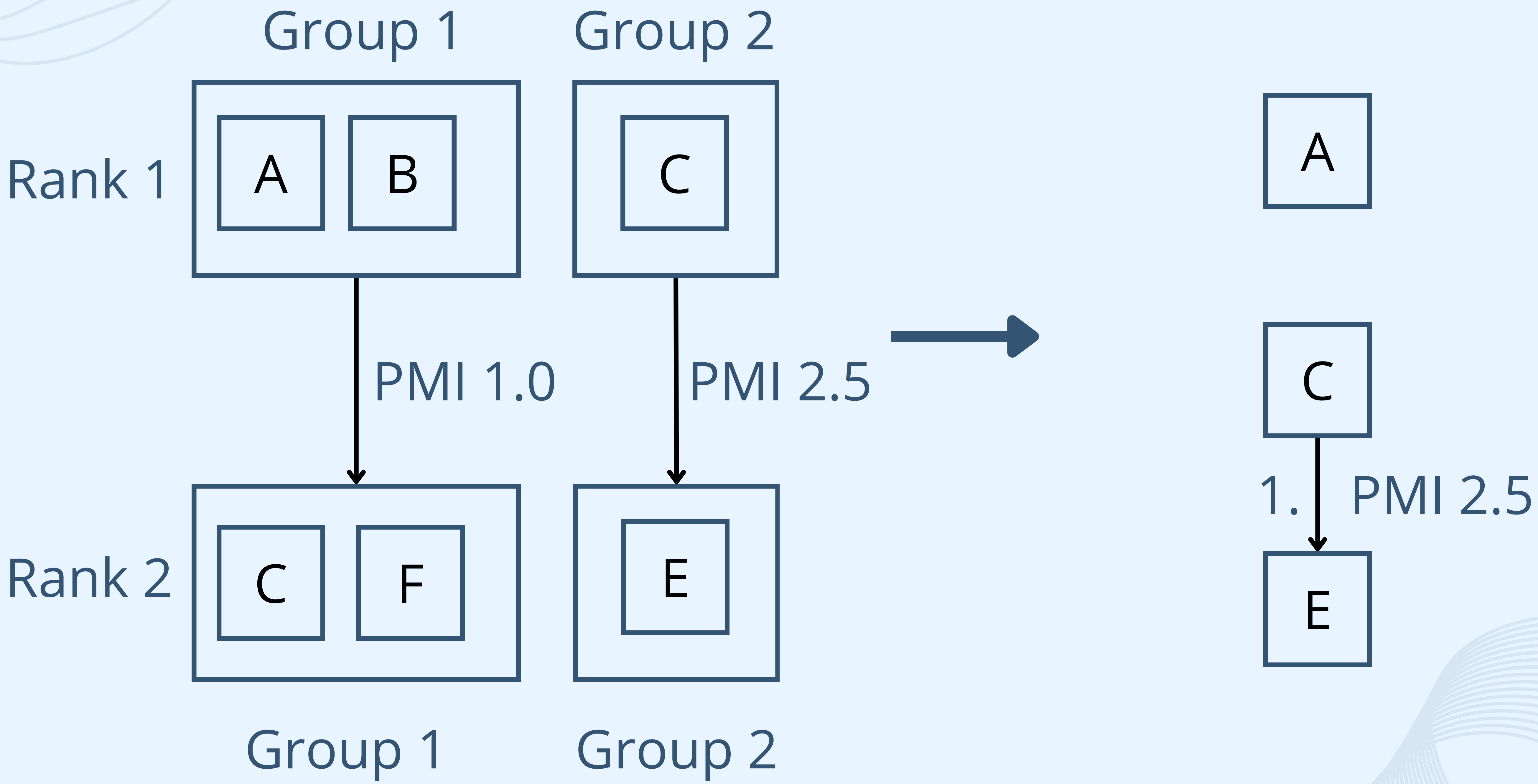
  - PMI $\leqslant$ 0: The events are independent or mutually exclusive. No edge is drawn.

- **Result**: The edges of the Directed Acyclic Graph (DAG) are now strictly defined and weighted, revealing the significant gene trajectories between the different gene subgroups

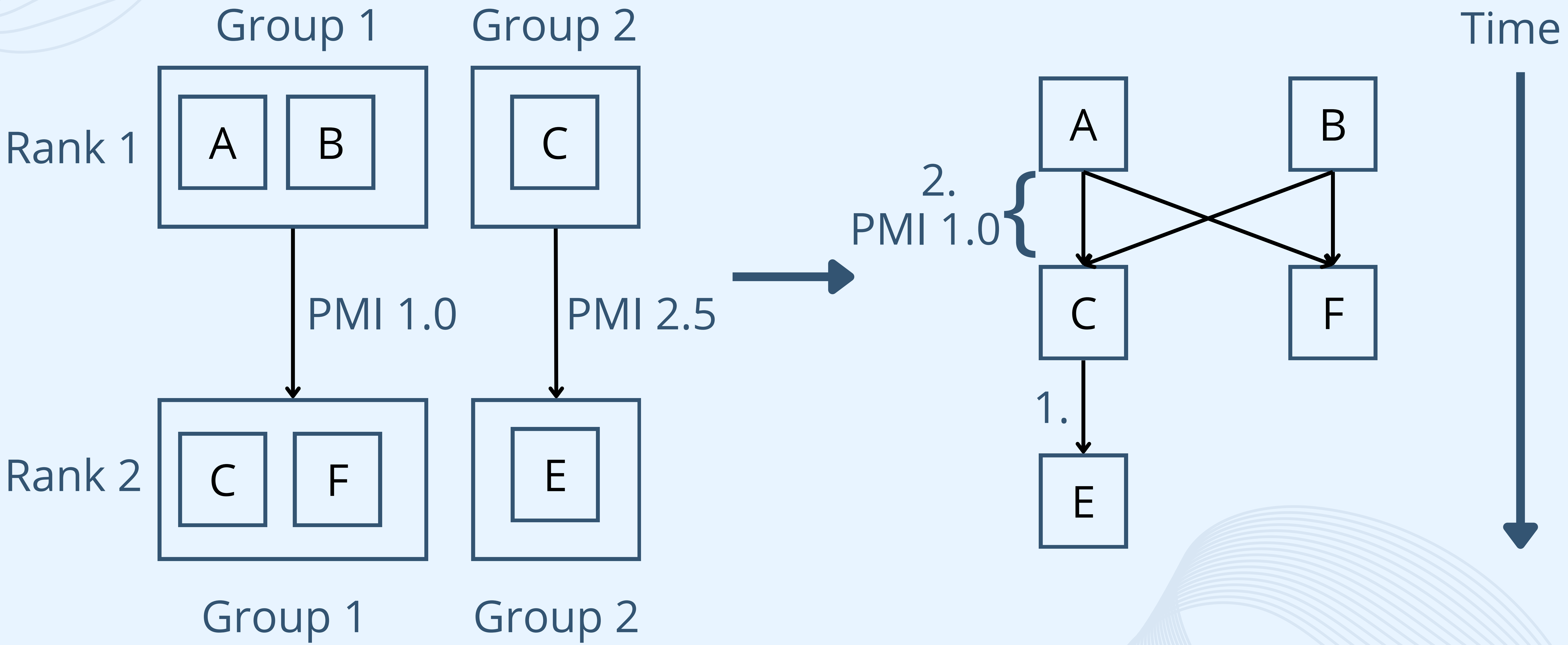
D. From Cluster trajectories to Single-Gene trajectories: Kruskal's Algorithm

- Transitions between clusters turn into transitions between genes
  - Example: AB to CD gives: A to C, A to D, B to C, B to D
- Treat genes as nodes and transitions as edges to form a graph
  - This lets us map how genes connect
- Order the edges by pmi from high to low
- Add each edge one at a time from the top of the list
- Skip any edge that would create a loop
- The result is a clean directed acyclic graph

D. From Cluster trajectories to Single-Gene trajectories: EXAMPLE



D. From Cluster trajectories to Single-Gene trajectories: EXAMPLE



RESULTS

- **Methodological Approach**: From Single Tumor to Merged Network
  - To preserve the specific biological context, the analysis and clustering pipeline was first applied at the **single tumor-type** level;
  - Subsequently, the **intra-tumor trajectories** were evaluated and merged to extract the final DAG.

- Outputs DAGs:
  - The following DAGs represent the evolutionary progression extracted from the ESAD (Esophageal Adenocarcinoma) cohort:
    - <u>Patients-level clustering</u>: rank connection
    - <u>Genes-level clustering</u>: rank connection
    - <u>Genes trajectories</u>: PMI
  - Final merge DAG: all the tumors

THE GRAPH

04 CONCLUSIONS

- **Biological Validation**:
The extracted trajectories reflect real biological events, showing high consistency with clinical data.
    - *Example*: Early driver genes like TP53 and PIK3CA are correctly placed at the far left as root nodes, while later-stage genes like ERBB2 are positioned in the middle, matching known clinical trajectories.

- **Clinical Utility**:
    - Predictive oncology: Identifying a patient's current position on this graph allows clinicians to statistically forecast the tumor's future evolutionary steps.
    - Finding therapy targets: Tracing the DAG backward reveals the early drivers of complex cascades, providing potential targets to stop the tumor progression.

THANKS